Roll No._____

DRONACHARYA COLLEGE OF ENGINEERING, GURUGRAM
PRE-UNIVERSITY EXAMINATION APRIL, 2025
(Gurugram University)

Semester: IV (B.Tech)

Branch: CSE (AI&ML)

Programming in Java

Subject with Code: Programming in Java

Time: 3 hours **Max. Marks: 70**

Note: *Q.1 is compulsory. Attempt any ONE question from each Unit. Each question carries of 14 marks. Assume data, if any missing.*

Q. 1. Short-answer type questions. **(2x7=14)**

i. What is Final keyword? (CO2)
ii. What is meant by static and dynamic binding? (CO2)
iii. What is Thread Priority? (CO3)
iv. What is JAR format? (CO1)
v. What is JVM and JRE? (CO1)
vi. Differentiate Enumeration and Iterator (CO4)
vii. What are checked exceptions in JAVA? (CO3)

UNIT-I

Q. 2 **(a)** What is a sandbox? Describe the Java sandbox security model. **(CO1)**
(b) What is Java Naming Convention? What are the advantages of naming conventions in Java? **(CO1)**

Q. 3 **(a)** What is Array? How is it initialized in java? Write a program to find the transpose of Matrix. **(CO1)**
(b) Explain any five Java features briefly. **(CO1)**

UNIT-II

Q. 4 **(a)** Elucidate constructor in Java and how is it implemented? Write an example to demonstrate constructor overloading? **(CO2)**
(b) What does the term "applet" refer to? Elaborate the life cycle of an applet? **(CO2)**

Q. 5(a) Define inheritance and discuss the different types of inheritance. Provide a Java program illustrating Multilevel inheritance. **(CO2)**
(b) Explain the Constructor chaining with the help of example. What are the rules and needs of constructor chaining? **(CO2)**

UNIT-III

Q. 6 **(a)** Explain the following terms with respect to exception handling. **(CO3)**
i) try **ii)** catch **iii)** throw **iv)** finally
(b) What role do the wait, notify, and notify All methods play in Java and why are they significant? **(CO3)**

Q.7 **(a)** Explain the importance of thread synchronization in preventing race conditions in concurrent programs. Discuss synchronized blocks and methods as mechanisms for achieving synchronization, along with their implications for shared resource consistency and performance. **(CO3)**

(b) Define Package. Explain various types Package in Java **(CO3)**

UNIT-IV

Q. 8 **(a)** What is Framework? What are the advantages of Collection Framework? **(CO4)**

(b) What is JDBC? Explain the steps for connectivity between Java program and database. **(CO4)**

Q. 9 **(a)** What is the significance of layout managers? Discuss about various layout managers. **(CO4)**

(b) What is interface? What is the major difference and similarity between interface and class? **(CO2)**